

● HARD

● ADVANCED MATH

● ANSWER KEY

Nonlinear equations in one variable and systems of equations in two variables

01

10 answers · Rationale-rich · Teach from doc

Q1**50**

The correct answer is 50. An equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has no real solutions if and only if its discriminant, $b^2 - 4ac$, is negative. Applying the distributive property to the left-hand side of the equation $kx^2 - 56x + 16 = 0$ yields $kx^2 - 56x + 16 = 0$. Adding 16 to each side of this equation yields $kx^2 - 56x + 16 = 0$. Substituting k for a , -56 for b , and 16 for c in $b^2 - 4ac$ yields a discriminant of $-56^2 - 4k16$, or $3,136 - 64k$. If the given equation has no real solution, it follows that the value of $3,136 - 64k$ must be negative. Therefore, $3,136 - 64k < 0$. Adding $64k$ to both sides of this inequality yields $3,136 < 64k$. Dividing both sides of this inequality by 64 yields $49 < k$, or $k > 49$. Since it's given that k is an integer, the least possible value of k is 50.

Q2**29**

The correct answer is 29. The solutions to a quadratic equation in the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be calculated using the quadratic formula, which gives two solutions: $x = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$. The given equation is in the form $ax^2 + bx + c = 0$, where $a = 1$, $b = 7$, and $c = 5$. It follows that one solution to the given quadratic equation is $x = \frac{-7 + \sqrt{7^2 - 4(1)(5)}}{2(1)}$, or $x = \frac{-7 + \sqrt{49 - 20}}{2}$, which is equivalent to $x = \frac{-7 + \sqrt{29}}{2}$. It's given that one solution to the given equation can be written as $x = \frac{-7 + \sqrt{k}}{2}$, where k is a constant. It follows that the value of k is 29.

Q3**D**

D. $-\sqrt{c^2 + 39^2}$

Choice D is correct. If $x^2 - c^2 \leq 0$, then neither side of the given equation is defined and there can be no solution. Therefore, $x^2 - c^2 > 0$. Subtracting $\frac{c^2}{\sqrt{x^2 - c^2}}$ from both sides of the given equation yields $\frac{x^2}{\sqrt{x^2 - c^2}} - \frac{c^2}{\sqrt{x^2 - c^2}} = 39$, or $\frac{x^2 - c^2}{\sqrt{x^2 - c^2}} = 39$. Squaring both sides of this equation yields $\frac{(x^2 - c^2)^2}{\sqrt{x^2 - c^2}^2} = 39^2$, or $\frac{x^2 - c^2}{x^2 - c^2} = 39^2$. Since $x^2 - c^2$ is positive and, therefore, nonzero, the expression $\frac{x^2 - c^2}{x^2 - c^2}$ is defined and equivalent to 1. It follows that the equation $\frac{x^2 - c^2}{x^2 - c^2} = 39^2$ can be rewritten as $\frac{x^2 - c^2}{x^2 - c^2} x^2 - c^2 = 39^2$, or $1x^2 - c^2 = 39^2$, which is equivalent to $x^2 - c^2 = 39^2$. Adding c^2 to both sides of this equation yields $x^2 = c^2 + 39^2$. Taking the square root of both sides of this equation yields two solutions: $x = \sqrt{c^2 + 39^2}$ and $x = -\sqrt{c^2 + 39^2}$. Therefore, of the given choices, $-\sqrt{c^2 + 39^2}$ is one of the solutions to the given equation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q4**C**

C. $\frac{prs}{pr + ps + rs}$

Choice C is correct. Multiplying each side of the given equation by $\frac{1}{20}$ yields $\frac{1}{20} \frac{20}{p} = \frac{1}{20} \frac{20}{q} - \frac{20}{r} - \frac{20}{s}$. Distributing $\frac{1}{20}$ on each side of this equation yields $\frac{20}{20p} = \frac{20}{20q} - \frac{20}{20r} - \frac{20}{20s}$, or $\frac{1}{p} = \frac{1}{q} - \frac{1}{r} - \frac{1}{s}$. Adding $\frac{1}{r} + \frac{1}{s}$ to each side of this equation yields $\frac{1}{p} + \frac{1}{r} + \frac{1}{s} = \frac{1}{q}$. Multiplying $\frac{1}{p}$ by $\frac{pr}{pr}$, $\frac{1}{r}$ by $\frac{rs}{rs}$, and $\frac{1}{s}$ by $\frac{rs}{rs}$ yields $\frac{pr}{prs} + \frac{ps}{prs} + \frac{rs}{prs} = \frac{1}{q}$, which is equivalent to $\frac{pr + ps + rs}{prs} = \frac{1}{q}$. Since $\frac{pr + ps + rs}{prs} = \frac{1}{q}$, and it's given that p , q , r , and s are positive, it follows that the reciprocals of each side of this equation are also equal. Thus, $\frac{prs}{pr + ps + rs} = \frac{q}{1}$, or $\frac{prs}{pr + ps + rs} = q$. Therefore, $\frac{prs}{pr + ps + rs}$ is equivalent to q .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5**29/3**

Accept also: 9.666, 9.667

The correct answer is $\frac{29}{3}$. Applying the distributive property to the left-hand side of the given equation, $xx + 1 - 56$, yields $x^2 + x - 56$. Applying the distributive property to the right-hand side of the given equation, $4xx - 7$, yields $4x^2 - 28x$. Thus, the equation becomes $x^2 + x - 56 = 4x^2 - 28x$. Combining like terms on the left- and right-hand sides of this equation yields $0 = 4x^2 - x^2 + -28x - x + 56$, or $3x^2 - 29x + 56 = 0$. For a quadratic equation in the form $ax^2 + bx + c = 0$, where a , b , and c are constants, the quadratic formula gives the solutions to the equation in the form $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Substituting 3 for a , -29 for b , and 56 for c from the equation $3x^2 - 29x + 56 = 0$ into the quadratic formula yields $x = \frac{29 \pm \sqrt{(-29)^2 - 4(3)(56)}}{2(3)}$, or $x = \frac{29}{6} \pm \frac{13}{6}$. It follows that the solutions to the given equation are $\frac{29}{6} + \frac{13}{6}$ and $\frac{29}{6} - \frac{13}{6}$. Adding these two solutions gives the sum of the solutions: $\frac{29}{6} + \frac{13}{6} + \frac{29}{6} - \frac{13}{6}$, which is equivalent to $\frac{29}{6} + \frac{29}{6}$, or $\frac{29}{3}$. Note that 29/3, 9.666, and 9.667 are examples of ways to enter a correct answer.

Q6**5**

The correct answer is 5. The intersection points of the graphs of $y = 3x^2 - 14x$ and $y = x$ can be found by solving the system consisting of these two equations. To solve the system, substitute x for y in the first equation. This gives $x = 3x^2 - 14x$. Subtracting x from both sides of the equation gives $0 = 3x^2 - 15x$. Factoring $3x$ out of each term on the left-hand side of the equation gives $0 = 3x(x - 5)$. Therefore, the possible values for x are 0 and 5. Since $y = x$, the two intersection points are $(0,0)$ and $(5,5)$. Therefore, $a = 5$.

Q7**C**

C. $w = \frac{x^2}{y} - 19$

Choice C is correct. Dividing each side of the given equation by 2 yields $\frac{14x}{14y} = \frac{2\sqrt{w+19}}{2}$, or $\frac{x}{y} = \sqrt{w+19}$. Because it's given that each of the variables is positive, squaring each side of this equation yields the equivalent equation $\frac{x^2}{y} = w + 19$. Subtracting 19 from each side of this equation yields $\frac{x^2}{y} - 19 = w$, or $w = \frac{x^2}{y} - 19$.

Choice A is incorrect. This equation isn't equivalent to the given equation.

Choice B is incorrect. This equation isn't equivalent to the given equation.

Choice D is incorrect. This equation isn't equivalent to the given equation.

Q8**C**

C. -25

Choice C is correct. It's given that the equation $-9x^2 + 30x + c = 0$ has exactly one solution. A quadratic equation of the form $ax^2 + bx + c = 0$ has exactly one solution if and only if its discriminant, $-4ac + b^2$, is equal to zero. It follows that for the given equation, $a = -9$ and $b = 30$. Substituting -9 for a and 30 for b into $b^2 - 4ac$ yields $30^2 - 4(-9)c$, or $900 + 36c$. Since the discriminant must equal zero, $900 + 36c = 0$. Subtracting $36c$ from both sides of this equation yields $900 = -36c$. Dividing each side of this equation by -36 yields $-25 = c$. Therefore, the value of c is -25.

Choice A is incorrect. If the value of c is 3, this would yield a discriminant that is greater than zero. Therefore, the given equation would have two solutions, rather than exactly one solution.

Choice B is incorrect. If the value of c is 0, this would yield a discriminant that is greater than zero. Therefore, the given equation would have two solutions, rather than exactly one solution.

Choice D is incorrect. If the value of c is -53, this would yield a discriminant that is less than zero. Therefore, the given equation would have no real solutions, rather than exactly one solution.

Q9**51**

The correct answer is 51. A quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has no real solution if and only if its discriminant, $-4ac + b^2$, is negative. In the given equation, $a = -1$ and $c = -676$. Substituting -1 for a and -676 for c in this expression yields a discriminant of $b^2 - 4(-1)(-676)$, or $b^2 - 2,704$. Since this value must be negative, $b^2 - 2,704 < 0$, or $b^2 < 2,704$. Taking the positive square root of each side of this inequality yields $b < 52$. Since b is a positive integer, and the greatest integer less than 52 is 51, the greatest possible value of b is 51.

Q10**A**A. $\frac{1}{57}$

Choice A is correct. The left-hand side of the given equation is the expression $57x^2 + 57b + ax + ab$. Applying the distributive property to this expression yields $57x^2 + 57bx + ax + ab$. Since the first two terms of this expression have a common factor of $57x$ and the last two terms of this expression have a common factor of a , this expression can be rewritten as $57xx + b + ax + b$. Since the two terms of this expression have a common factor of $x + b$, it can be rewritten as $x + b57x + a$. Therefore, the given equation can be rewritten as $x + b57x + a = 0$. By the zero product property, it follows that $x + b = 0$ or $57x + a = 0$. Subtracting b from both sides of the equation $x + b = 0$ yields $x = -b$. Subtracting a from both sides of the equation $57x + a = 0$ yields $57x = -a$. Dividing both sides of this equation by 57 yields $x = \frac{-a}{57}$. Therefore, the solutions to the given equation are $-b$ and $\frac{-a}{57}$. It follows that the product of the solutions of the given equation is $-b\frac{-a}{57}$, or $\frac{ab}{57}$. It's given that the product of the solutions of the given equation is kab . It follows that $\frac{ab}{57} = kab$, which can also be written as $ab\frac{1}{57} = abk$. It's given that a and b are positive constants. Therefore, dividing both sides of the equation $ab\frac{1}{57} = abk$ by ab yields $\frac{1}{57} = k$. Thus, the value of k is $\frac{1}{57}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank